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Microgrid Frequency Regulation and Optimal Sizing of Emergency Generator Considering VSG Coupled Electric Vehicles

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Abstract—Recently, the transition from conventional to renewable energy sources (RESs), from internal combustion engine vehicles to electric vehicles (EVs), and from the main grid to microgrids (MGs) are essential goals to both reduce greenhouse gas emissions and ensure the stability of power systems. However, the transitions cause new concerns in the grid including technical challenges and financial viability. This study examines the RES-based MG under realistic conditions considering the uncertainty in fleet size of EVs, emergency generator capacity, solar irradiation, and wind speed in island mode. The work aims to provide effective solutions including conventional methods alongside today's trend namely doubly fed electrical generators (DFIG), vehicle-to-grid mechanism (V2G), maximum power point tracking controller (MPPT), voltage source inverter (VSI) with high switching frequency for technical challenges, and virtual synchronous generator (VSG) mechanism for financial viability. The paper provides a guide for resizing the emergency generators capacity depending on system instability. The observations verify that the control mechanisms reinforce the system to remain stable by decreasing the range of frequency fluctuation from 3.1 to <0.05 Hz, the peak point of frequency from 51.8 to 50.05 Hz, and the emergency generator capacity from 0.7605 to 0.3420 MVAr at MATLAB and Simulink.

Keywords: distributed renewable energy source, frequency regulation, microgrid, optimal emergency generator sizing, virtual synchronous generator
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1. INTRODUCTION

1.1. Background

One of the main causes of climate change is the increase in carbon dioxide emissions caused by the use of fossil fuels to meet increasing global energy demand [1]. Also, there is a relationship between climate change and increasing natural disaster rates [2]. Today, grid control and management methods must consider both the increasing demands and the possibility of natural disasters together. Therefore, the transition from conventional to renewable energy sources (RESs), from internal combustion engine

vehicles to electric vehicles (EVs), and from the main grid to microgrid (MG) are essential goals to decarbonize and protect the parts of power systems against natural disasters [2]. However, the trends in power systems have led to new sensitivities to keep the stability of power systems [3]. Therefore, the correlation between the increase in the rate of natural disasters, and the tendencies toward RESs and EVs require evaluation concurrently.

Natural disasters and cyber-attacks, which may cause temporary or long-term blackouts in grids, put the concept of MG forward to provide system reliability and stability due to the flexibility of the concept in the operational characteristic of the grid [2, 4]. The term MG is qualified to operate synchronously with the main grid in grid-tied mode and independently from the main grid in island mode or switch between the operation modes [5]. The supply-demand balance is stable in the grid-tied mode, in contrast to the volatility in the intermittent nature of RESs-based MGs in island mode [6]. In many works, RES-based distributed generations (DGs) have non-existent or low inertia characteristics and interface to the MG *via* a power electronic interface (PEI) [7]. The approach has both a shorter response time and easy controllability than conventional synchronous generators (SGs), but the systems have still suffered unstable inertia (*i.e.*, instability) [5]. This decrease results increase rate-of-change-of-frequency and frequency deviations under power mismatch [8]. Therefore, the virtual synchronous generator (VSG) mechanism, which mimics the dynamic character of SG by emulating the swing equation of SG to enhance the transient response of the system in real-time, has been proposed to overcome the deficiencies in literature [9, 10].

1.2. Literature Review

Many experts and researchers work on optimal MG control considering trends and sensitivities, thus emphasizing the importance of the VSG control mechanism for auxiliary frequency control (FC) by providing necessary inertia [11–13]. Dhingra *et al.* [14] proposed a VSG mechanism to provide demanded inertia for RES-based MG with a charging station (CS) consisting of a fleet of EVs considered as an ESS and verified that VSG-based CS responds effectively to dynamic load changes and environmental uncertainties in a shorter time in comparison with conventional methods for FC. In Ref. [15], a multi-CS structure connected to the common DC bus was introduced to increase system stability and efficiency. The model evaluated the state of charge, load demand, and solar irradiation level as independent variables, and frequency as the dependent variable, and examined the effects of the coordinated VSGs for FC. In Ref. [16], the importance of multiple VSG control with the same concerns in Ref. [15] was also emphasized, and drawn attention to the negative impacts of using centralized controllers for FC such as dependence on the communication system, and proposed a decentralized control mechanism. Li *et al.* [17] demonstrated the improvement in response time of the VSG mechanism in islanded-mode FC in comparison with conventional methods and highlighted the importance of the number of EVs involved in FC. Ref. [18] included JAYA optimization for frequency stabilization and conducts a comparative analysis of LFC designs. Suul *et al.* [19] examined the virtual synchronous machine-based control (VSM) mechanism of a single-phase charger in both time-domain simulations and laboratory experiments, and proposed a VSM mechanism as a suitable option in the use of Vehicle-to-Grid (V2G) with single-phase battery chargers. Ref. [20] emphasized Automatic Voltage Regulator (AVR) in maintaining a constant voltage level of SG and provides elasticity of dynamic response of this AVR scheme against disturbances. Yap *et al.* [21] investigated and simulated the impact of virtual inertia control strategies under variable solar irradiance and temperature through a Maximum Power Point Tracker (MPPT) controller and fault-ride through. Similarly, improvement in frequency regulation through the related control methods in comparison with conventional inverters and PI-based synconverter was also observed in Refs. [17, 21]. The VSG mechanism in a double-fed induction generator (DFIG) was examined in Ref. [22], and it depicted the contribution of the grid-connected wind power storage system providing virtual inertia to suppress frequency fluctuations. Ref. [23] examined the lack of inertia in the area including wind power plants and proves the effectiveness of automatic generation controller to classical Genetic algorithm-tuned controller.

1.3. Research Gap and Motivation

The characteristic remarks of the aforementioned studies were revealed in Table 1. Among all the studies that use the VSG mechanism, while Refs. [14–16, 19] did not include wind power generation, Refs. [16, 19, 22] did not contain solar energy production on the supply side. Among the studies containing solar generation, while Ref. [15] did not has a proper control mechanism, Refs. [17, 21] did not contain DFIG as a configuration contrary to our study. Finally, although Refs. [14, 15, 17, 19] comprised the VSG

	Components				Control and tech.				
	RES								
References	Solar	Wind	EV	SG	V2G	VSG	MPPT	DFIG	Design guide
[14]	✓	✗	✓	✓	✓	✓	✓	✗	✗
[15]	✓	✗	✓	✗	✓	✓	✗	✗	✗
[16]	✗	✗	✗	✗	✗	✓	✗	✗	✗
[17]	✓	✓	✓	✓	✓	✓	✓	✗	✗
[18]	✗	✗	✓	✗	✓	✓	✗	✗	✗
[19]	✓	✓	✗	✗	✗	✓	✓	✗	✗
[20]	✗	✓	✗	✗	✗	✓	✗	✓	✗
This paper	✓	✓	✓	✓	✓	✓	✓	✓	✓

TABLE 1. Comparison of the proposed algorithm with related literature works.

mechanism and the V2G concept, they did not contain a guide with respect to the variable size of the fleet based on V2G concept, which is discussed in terms of usage for FC in the paper, and the emergency generator capacity.

1.4. Challenges

External impacts such as cyber-attacks, natural events such as natural disasters, and uncertain behavior of RES-based DGs are some challenges for a resilient power system. In this regard, the new approaches with the challenges handled in this paper are stated from various aspects in this section. The paper examines dynamic changes in the power system stability depending on the frequency fluctuation in the MG when the power transmission was interrupted by external causes such as natural disasters or cyber-attacks. Moreover, it focuses that today's MGs mainly suffer from insufficient inertia due to the intermittent nature of RES-based DGs in the case of on-grid connection. The study aims to take advantage of EVs that are able to flow bidirectional power between the grid and vehicle battery. Therefore, the proposed CS is connected to the MG through a PEI, which is controlled by the VSG mechanism, to compensate for the insufficient inertia caused by power imbalances. The response time and characteristics of the components are monitored to demonstrate the superiority of the VSG mechanism over conventional systems in FC. In addition, the other purpose is to provide a guide for determining the optimal emergency generator capacity in MGs, based on the level of EV integration and various environmental variables.

1.5. Contributions

It is crucial to design and control an MG that efficiently performs power generation, distribution, and storage while minimizing interruptions and downtime, especially in island mode. The approach for the main problems might be analyzed in technical challenges and financial viability in this paper. The major contributions are listed as follows according to the literature gaps.

- Ensuring compatibility, optimizing power flow, managing voltage fluctuations, and maintaining grid stability, which is overcome by applying a combination of effective methods namely DFIG, MPPT, VSI, V2G, and VSG mechanism.
- In this study, the effect of instantaneous changes in the supply side consisting of a decrease in solar irradiation, wind speed, and the fleet size of EVs is visualized based on the active and reactive power characteristics of the system elements.
- In assessing the financial viability, the paper considers the efficient use of financing resources by guiding in the sense of rescaling of emergency SGs.

1.6. Paper Organization

The rest of this paper is organized as follows: Section 2 provides information about the components and their control methods, including their mathematical equivalents. Section 3 presents the main control approach and its representation in the control flow. In Section 4, the input data is presented, and the results are interpreted to validate the effectiveness of the proposed control mechanism. Finally, the conclusions are shared to guide the reader in determining the optimal emergency generator.

2. METHODOLOGY

2.1. Overview of the Structure

The schematic diagram of the proposed structure and subsections is shown in Figure 1. The structure is divided into different zones based on the operating characteristics of the units.

The study assumes that an external factor, such as a natural disaster or cyber-attack, forces the MG to operate in island mode, and the proposed control approaches aim to maintain stability in the power system. The modeled grid region has an energy management system (EMS) to ensure the internal stability of the system in the island mode. The system consists of an MPPT-controlled solar panel in 1st subsystem, an SG in 2nd subsystem worked as an emergency unit on the supply side, DFIG-based wind power in 3rd subsystem, and V2G-based charger point in 4th subsystem, while a residential load area shown as 5th subsystem is located on the demand side. When external factors force the system to operate in island mode, the aim of the EMS is to prevent power mismatches that cause system instability by continuously comparing the dynamic production and consumption values. The EMS regulates the operation of converters using the VSG mechanism based on the characteristics of the power mismatch. If the load side is dominant, the system operates the converter in boost mode, and if the demand side is dominant, the system operates the converter in buck mode. Furthermore, if the methodology is insufficient for regulating system frequency, the EMS will control the output power of the EG, as described in the following sections. Therefore, the EMS determines the characteristics of the charger and the output power value of the EG to ensure that system frequency occurs within the standard frequency range [24]. The scenario is tested based on the size of the EV fleet, which determines the control range, and the environmental factors that affect production characteristics. The output power of the EG and the system frequency are then analyzed.

2.2. Architectural Design of Charger Integration

The proposed model includes a bidirectional converter to take advantage of EVs that shall conveniently be operated on both the demand side as a load unit and the supply side as a generation unit. Thus, EVs charge as a load at buck mode while they discharge as a source at boost mode. The stability analysis of the converter is realized in the state-space model. The state space and the output equations of the buck–boost converter represent in (1) and (2):

$$x'_{1,2} = A_{1,2}.x + B_{1,2}.u \tag{1}$$

$$y_{1,2} = C_{1,2}.x + D_{1,2}.u \tag{2}$$

Where x' represents the differential form of the state variables when x represents the state variables considering the current through the converter's inductance (I_b), and the battery terminal voltage (V_{dc}), respectively. Also, y represents the output variable and u indicates the switch position function.

The proposed converter consists of a voltage source, inductor, capacitor, switch, and resistor as presented in Figure 2(a). The figure encompasses generic architecture and specific operating modes for a buck–boost converter. Kirchhoff's voltage and current laws are applied to the converter's closed

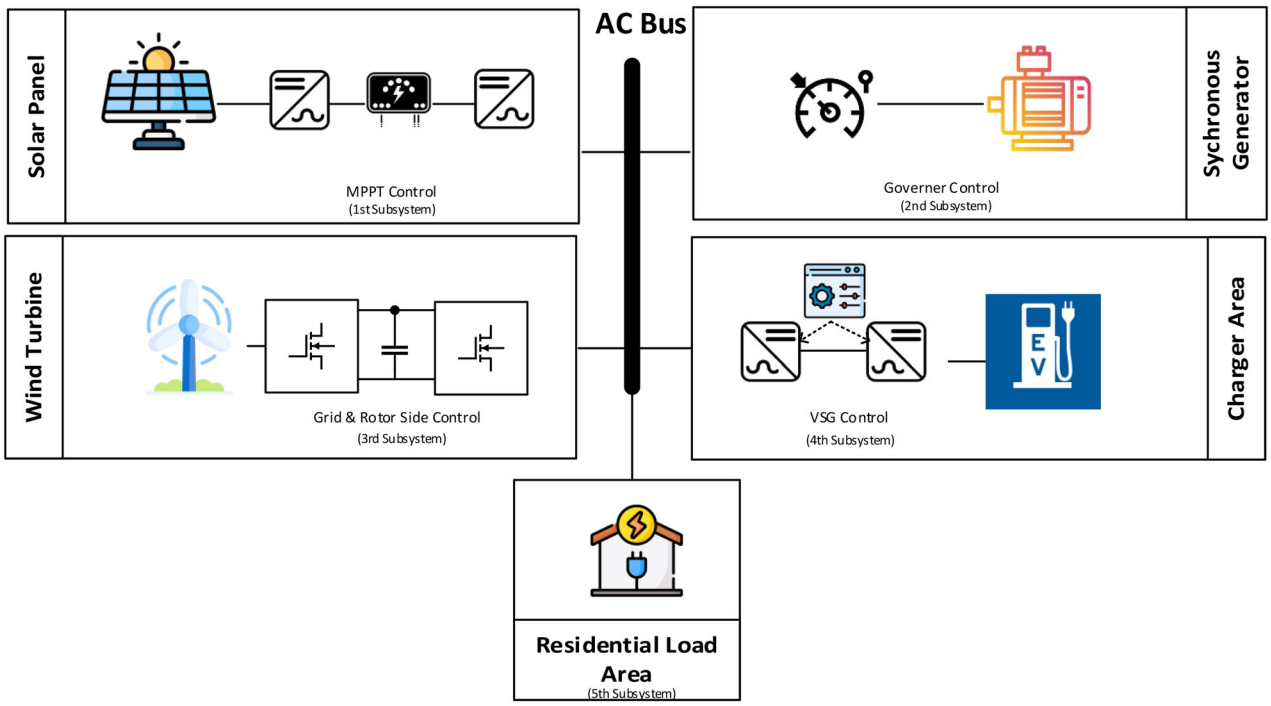


FIGURE 1. The schematic diagram of the proposed structure.

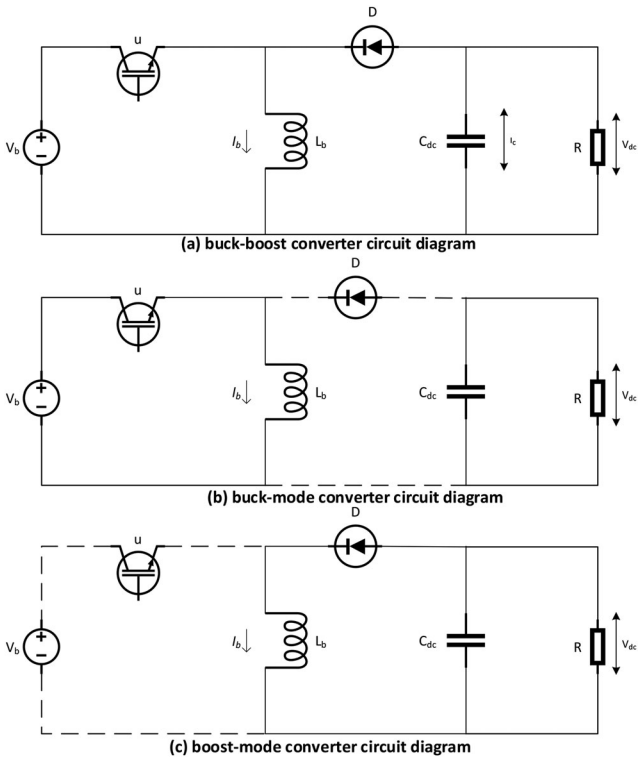


FIGURE 2. The schematic diagram of the buck–boost converter. (a) Buck–boost converter circuit diagram. (b) Buck-mode converter circuit diagram. (c) Boost-mode converter circuit diagram.

circuits at buck mode, respectively as demonstrated in Figure 2(b) and formulated as in (3) and (4):

$$V_b - L_b.\frac{dI_b}{d_t} = 0 \tag{3}$$

$$\frac{V_{dc}}{R} + C_{dc}.\frac{dV_{dc}}{d_t} = 0 \tag{4}$$

The matrices of the converter at buck mode are written in (5) and (6). A_1, B_1, C_1, and D_1 are the $ABCD$ state space parameters of the buck–boost converter at buck mode with respect to Eqs. (1) and (2).

$$\begin{aligned} x_1' = \begin{bmatrix} \frac{dI_b}{d_t} \\ \frac{dV_{dc}}{d_t} \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & \frac{-1}{R.C_{dc}} \end{bmatrix}.\begin{bmatrix} I_b \\ V_{dc} \end{bmatrix} + \begin{bmatrix} \frac{1}{L_b} \\ 0 \end{bmatrix}.V_b \\ \left.\begin{bmatrix} 0 & 0 \\ 0 & \frac{-1}{R.C_{dc}} \end{bmatrix}\right\}A_1, \left.\begin{bmatrix} \frac{1}{L_b} \\ 0 \end{bmatrix}\right\}B_1 \end{aligned} \tag{5}$$

$$\begin{aligned} y_1 = \begin{bmatrix} 0 & 1 \end{bmatrix}.\begin{bmatrix} I_b \\ V_{dc} \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix}.V_b \\ \left.\begin{bmatrix} 0 & 1 \end{bmatrix}\right\}C_1, \left.\begin{bmatrix} 0 \\ 0 \end{bmatrix}\right\}D_1 \end{aligned} \tag{6}$$

Kirchhoff's voltage and current laws are applied to the converter's closed circuits at boost mode, respectively as demonstrated in Figure 2(c), and formulated as in (7) and (8):

$$L_b.\frac{dI_b}{d_t} + V_{dc} = 0 \tag{7}$$

$$I_b - \frac{V_{dc}}{R} - C_{dc}.\frac{dV_{dc}}{d_t} = 0 \tag{8}$$

The matrices of the converter at boost mode are written in (9) and (10). A_2, B_2, C_2, and D_2 are the $ABCD$ state space parameters of the buck–boost converter at boost mode with respect to Eqs. (1) and (2).

$$\begin{aligned} x_2' = \begin{bmatrix} \frac{dI_b}{d_t} \\ \frac{dV_{dc}}{d_t} \end{bmatrix} = \begin{bmatrix} 0 & \frac{-1}{L_b} \\ \frac{1}{C_{dc}} & \frac{-1}{R.C_{dc}} \end{bmatrix}.\begin{bmatrix} I_b \\ V_{dc} \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix}.V_b \\ \left.\begin{bmatrix} 0 & \frac{-1}{L_b} \\ \frac{1}{C_{dc}} & \frac{-1}{R.C_{dc}} \end{bmatrix}\right\}A_2, \left.\begin{bmatrix} 0 \\ 0 \end{bmatrix}\right\}B_2 \end{aligned} \tag{9}$$

$$\begin{aligned} y_2 = \begin{bmatrix} 0 & 1 \end{bmatrix}.\begin{bmatrix} I_b \\ V_{dc} \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix}.V_b \\ \left.\begin{bmatrix} 0 & 1 \end{bmatrix}\right\}C_2, \left.\begin{bmatrix} 0 \\ 0 \end{bmatrix}\right\}D_2 \end{aligned} \tag{10}$$

Equations (11) and (12) represent the average state space model and the output equation of buck–boost converter. Furthermore, the matrices of the average parameters are written in (13) and (14). A, B, C, and D are the $ABCD$ parameters of the buck–boost converter with respect to Eqs. (11) and (12). Finally, in (15) and (16), the values of inductor and capacitor for the buck–boost converter are estimated.

$$x' = dx_1' + (1-d)x_2' \tag{11}$$

$$y = dy_1 + (1-d)y_2 \tag{12}$$

$$\begin{aligned} x' = \begin{bmatrix} \frac{dI_b}{d_t} \\ \frac{dV_{dc}}{d_t} \end{bmatrix} = \begin{bmatrix} 0 & \frac{d-1}{L_b} \\ \frac{1-d}{C_{dc}} & \frac{-1}{R.C_{dc}} \end{bmatrix}.\begin{bmatrix} I_b \\ V_{dc} \end{bmatrix} + \begin{bmatrix} \frac{d}{L_b} \\ 0 \end{bmatrix}.V_b \\ \left.\begin{bmatrix} 0 & \frac{d-1}{L_b} \\ \frac{1-d}{C_{dc}} & \frac{-1}{R.C_{dc}} \end{bmatrix}\right\}A, \left.\begin{bmatrix} \frac{d}{L_b} \\ 0 \end{bmatrix}\right\}B \end{aligned} \tag{13}$$

$$y = \begin{bmatrix} 0 & 1 \end{bmatrix}.\begin{bmatrix} I_c \\ V_{dc} \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix}.V_b \quad \begin{bmatrix} 0 & 1 \end{bmatrix}\}C, \begin{bmatrix} 0 \\ 0 \end{bmatrix}\Big\}D \tag{14}$$

$$L_b = \frac{(V_b + V_{dc}).\lambda.(1-d)}{\Delta I_b.f_s} \tag{15}$$

$$C_{dc} = \frac{V_{dc}.\lambda}{R.f_s.\Delta V_{dc}} \tag{16}$$

Where V_b and V_{dc} are considered as 400 V input voltage, and 1000 V output voltage, respectively. ΔI_b is the ripple current of the inductor considered as 1% of the actual inductor current. ΔV_{dc} is the ripple in output voltage considered as 5% of the actual output voltage. R is the resistance at the output terminal. λ is the duty cycle considered as 0.6, and f_s is the switching frequency considered as 20 kHz. Based on these parameters, L_b and C_{dc} are calculated as 172 μH and 881 μF, respectively.

2.3. Control Configuration of Charger Integration

The buck–boost converter controls the bi-directional power flow between the CS and MG by maintaining a constant voltage on the common DC bus. The reference voltage ($V_{bus,ref}$) for the DC-bus is compared with the actual DC-bus voltage (V_{bus}) to determine the reference current (I_{ref}) by a PI controller as depicted in Figure 3(a). Furthermore, the switching signal for the Pulse-width modulation (PWM) is determined by comparing the reference current (I_{ref}) with the actual current (I_{actual}) to switch the semiconductor as shown in Figure 3(b).

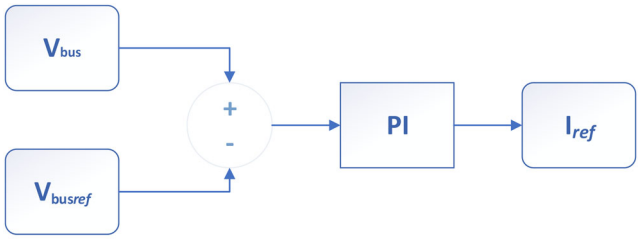
In (17) and (18), the reference current and voltage are formulated, respectively. The sensitivity of the controller is determined by the parameters of the PI controllers, proportional (K_{p^1,p^2}), and integral coefficients (K_{i^1,i^2}).

$$I_{ref} = (V_{bus} - V_{bus,ref}).\frac{K_{p^1}.s + K_{i^1}}{s} \tag{17}$$

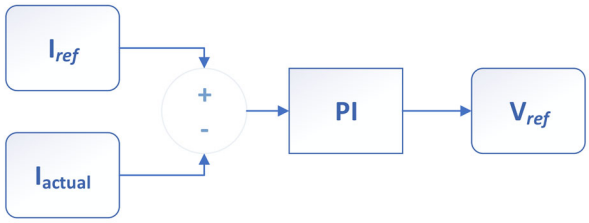
$$V_{ref} = (I_{ref} - I_{actual}).\frac{K_{p^2}.s + K_{i^2}}{s} \tag{18}$$

2.4. Control Configuration of Solar System

One of the aims of the paper is to observe and reveal the effect of solar variables on the supply side. Specifically, it focuses on the within-period fluctuation of solar irradiance



(a) Current control diagram



(b) Voltage control diagram

FIGURE 3. The control diagrams of the converter. (a) Current control diagram. (b) Voltage control diagram.

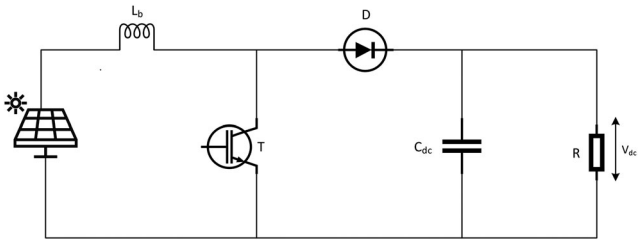


FIGURE 4. The generic diagram of the MPPT-controlled solar model.

in the renewable distributed generation that is controlled by the MPPT controller. The proposed solar system is modeled with a solar panel, inductor, capacitor, diode, switch, and resistor as depicted in Figure 4. The fundamental equations for Incremental Conductance-based MPPT are presented in (19) and (20).

$$\frac{dP}{dV} = \frac{d(V.I)}{dV} \tag{19}$$

$$\frac{dI}{dV} = -\frac{I}{V} \tag{20}$$

Where P is an active power, and V and I are the panel output voltage and current, respectively. The aim of the MPPT method is to produce maximum power depending on the dynamic conditions. Furthermore, the control diagram of the MPPT is shown in Figure 5. While I_{pv} and P_{pv} are solar panel output current and power, respectively, I'_{pv} and P'_{pv} are solar panel output current and power in the

previous step, respectively. The aim of the control diagram is to identify how the controller determines the input of the PWM. The controller compares the difference in solar panel output current and power dynamically to decide whether the duty cycle is lower or higher than the maximum power point with an accuracy of 0.001. The control method utilizes the derivative of power with respect to voltage as a variable. In (21)–(23), the system uses the relationship between the instantaneous conductance (I/V) and the incremental conductance ($\frac{dI}{dV}$) as formulated in Ref. [25].

$$\frac{dP}{dV} = 0,\ P \cong MPP,\ \text{and}\ \frac{dI}{dV} = -\frac{I}{V} \tag{21}$$

$$\frac{dI}{dV} > -\frac{I}{V},\ \text{at the left of MPP} \tag{22}$$

$$\frac{dI}{dV} < -\frac{I}{V},\ \text{at the right of MPP} \tag{23}$$

The controller monitors the change in the panel current and voltage toward the MPPT, where the ($\frac{dP}{dV}$) value converges to zero, as shown in (21), but the condition becomes rare. When ($\frac{dP}{dV}$) is negative as formulated in (22), the operation point of the Photovoltaic (PV) module is at the left of MPP, and the reference voltage has to be increased. Contrary, when ($\frac{dP}{dV}$) is positive as formulated in (23), the operation point of the PV module is at the right of MPP, and the reference voltage has to be decreased.

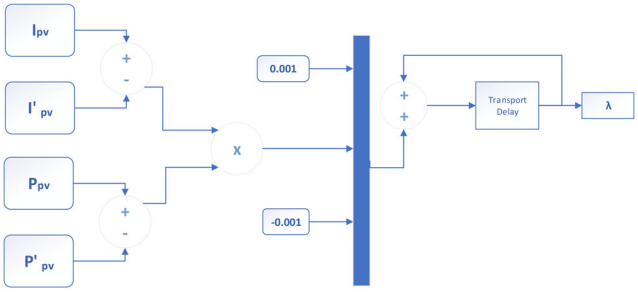


FIGURE 5. The generic diagram of the MPPT-controlled solar model.

2.5. Control Configuration of Doubly-Fed Induction Generator (DFIG) Wind Turbine

In our study, the source-side indecision is intentionally increased to investigate the VSG mechanism under realistic conditions. For this purpose, the performance of the solar system under varying irradiation levels is analyzed, as previously mentioned. Additionally, the operating modes of wind turbines under changing wind speeds are studied in the paper. In many studies, double-fed induction generators, which are visualized in Figure 6, are included to prevent the impact of sudden wind speed fluctuations on system components.

In DFIGs, asynchronous motors are employed to synchronize the fluctuations in wind speed with the stator frequency, which is equal to the grid frequency. To accomplish this, the stator windings are connected directly to the grid, while the rotor windings are connected to the grid through a bidirectional back-to-back converter. The rotor side converter compensates the impact of changes in wind speed on the rotor windings by utilizing active and passive power control, while the grid side converter maintains the constant dc-bus voltage [26]. The mechanical power and the stator electric power output are calculated in (24).

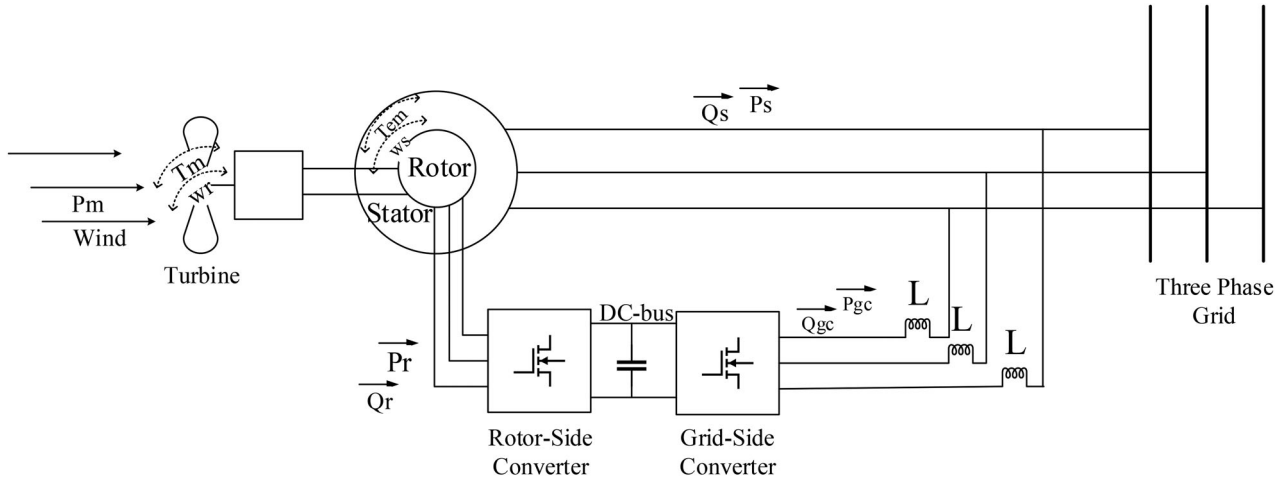


FIGURE 6. The generic diagram of the DFIG-based wind turbine model.

$$P_m = T_m w_r P_s = T_{em} w_s \tag{24}$$

While P_m and P_s represent the wind power and the stator electrical output power, respectively, T_m and T_{em} are the mechanical torque applied to the rotor by the turbine and the electromagnetic torque applied to the rotor by the generator, respectively. Furthermore, w_r and w_s state the rotor rotational speed and the magnetic flux rotational speed, respectively. When it is assumed that J is the combined rotor and wind turbine inertia coefficient, the difference of torque can be calculated in (25) assuming the equalities in steady state at fixed speed for a lossless generator given in (26) and (27).

$$J\frac{dw_r}{dt} = T_m - T_{em} \tag{25}$$

$$T_m = T_{em} \tag{26}$$

$$P_m = P_s + P_r \tag{27}$$

Equation (28) states the rotor electrical output power as derived from (24). s is defined in (29) as the slip of the generator. Furthermore, the rearranged form of (28) is presented in (30)

$$P_r = T_m w_r - T_{em} w_s \tag{28}$$

$$s = \frac{w_s - w_r}{w_s} \tag{29}$$

$$P_r = -sT_m w_s \tag{30}$$

It can be established the relationship between the rotor electrical output power and the stator electrical output power with respect to slip. In the model, wind speed is assumed as a variable to investigate its impact on power equality during transitions. The DFIG-based wind turbines have the capability of generating 1.6 MVAr at 15 m/s, but the conclusion section assesses their performance at a wind speed of 10 m/s.

2.6. Control Configuration of Emergency Generator

The integration of synchronous generators, referred to as emergency units, of varying sizes into MGs is crucial for the security of the system, especially RESs-based MGs. The generator capacity size has a direct impact on the control methods and operating characteristics of the system. In this study, different environmental conditions and control methods are considered, and the power values expected from the synchronous generator under various scenarios are taken into account and compared.

A typical diesel generator consists of a diesel engine governor, an alternator, and an excitation system, which includes a voltage regulator and an exciter. The voltage regulator is responsible for controlling the rotor output voltage, while the exciter provides field current to the rotor windings to generate the rotor output voltage [27]. The dynamic equations are presented in (31)–(33):

$$T_e\frac{dE_{fv}}{dt} = V_p - K_e E_{fv} \tag{31}$$

$$T_f\frac{dR_d}{dt} = -R_d + \frac{K_g}{T_f}E_{fv} \tag{32}$$

$$T_a\frac{dV_p}{dt} = -V_p + K_c R_d - \frac{K_c K_g}{T_f}E_{fv} + K_c(V_{ref} - V) \tag{33}$$

T_e, T_f, and T_a represent the time constant of the exciter, stabilizer, and amplifier, respectively. K_e, K_g, and K_c are the gain constant of the exciter, the self, and the amplifier, respectively. V_P, V_{ref}, V, and E_{fv} denote the output voltage of the pilot exciter, reference, diesel generator, and field, respectively. The term R_d represents the system rate feedback. Our study includes a MATLAB model based on Eqs. (31)–(33), and the response of the model to changes during simulation is analyzed in the conclusion section.

2.7. Control Configuration of Virtual Synchronous Generator (VSG)

The VSG control technique for the active power-frequency droop control method is used in many studies due to the simplicity to improve the dynamic characteristic of systems. Basically, although the aim of the VSG mechanism is to mimic the dynamic characteristic of a synchronous generator, it is fast and more flexible than the conventional method. This impact is important especially for island mode MGs, since the frequency can be stabilized as soon as possible. The VSG mechanism computes the virtual rotor angle (θ_r) to rearrange the relevant converter output power (P_{out}). Thus, the VSG mechanism compensates for the power mismatches and then, improves the system frequency. The simple swing equation of VSG is stated in Eq. (34), and the inertia moment and frequency tracking error are presented in Eqs. (35) and (36), respectively.

$$P_{ref} - P_{out} = \Delta P = Jw_r\left(\frac{dw_r}{dt}\right) + K\Delta w_r \tag{34}$$

$$J = \frac{2H}{w_o^2} S_{rated} \tag{35}$$

$$\Delta w_r = w_o - w_r \tag{36}$$

While P_{ref} and P_{out} represent the reference power and the output power, respectively, ΔP is the power mismatch. J is the inertia moment in kgm_2, K is a damping factor, H is the inertia constant, and S_b is the rated apparent power. Δw_r reflects the frequency tracking error when w_o and w_r are the nominal angular frequency and virtual angular frequency, respectively. In this regard, $\frac{dw_r}{dt}$ shows the change in virtual angular frequency which can be used for computing the virtual rotor angle (δ_r) as stated in (37).

$$\delta_r = \int_0^t w_r dt \tag{37}$$

The control diagram of the VSG is shown in Figure 7. Firstly, the inverter output voltage is used for handling the nominal angular frequency in the grid through phase-locked loop method in Figure 7(a). Similarly, the inverter output current (I_{abc}) is transformed from abc-domain to dq frame (I_{dqo}) to compare the reference current calculated ($I_{dq_{ref}}$) by buck–boost converter controller in Figure 7(a). The results are considered as input in dq frame and evaluated with the virtual rotor angle (δ_r) calculated in Figure 7(b) by comparing the reference power and actual power with the nominal angular frequency (w_r). Finally, PWM signal is received in Figure 7(c).

$$V_{d_{ref}} = (I_{d_{ref}} - I_d)\frac{K_{p^3}.s + K_{i^3}}{s} - wL_s I_q + V_d \tag{38}$$

$$V_{q_{ref}} = (I_{q_{ref}} - I_q)\frac{K_{p^4}.s + K_{i^4}}{s} - wL_s I_d + V_q \tag{39}$$

The control steps are formulated in (38) and (39). wL_s is the cross-coupling factor, and proportional (K_{p^3}, K_{p^4}) and integral coefficients (K_{i^3}, K_{i^4}) are the proportional and the integral coefficients for PI controller, respectively.

The VSG mechanism improves the overall stability and reliability of the power system by rapid response against transient events and emulating the inertia of synchronous generators. The VSG mechanism helps balance supply and demand for keeping grid frequency stable under transient events. Moreover, the VSG mechanism allows RESs to connect the system without causing low inertia, so the mechanism also supports MGs in a financial way in constant to conventional methods. Finally, the control method uses the frequency loop for frequency regulation. The VSG

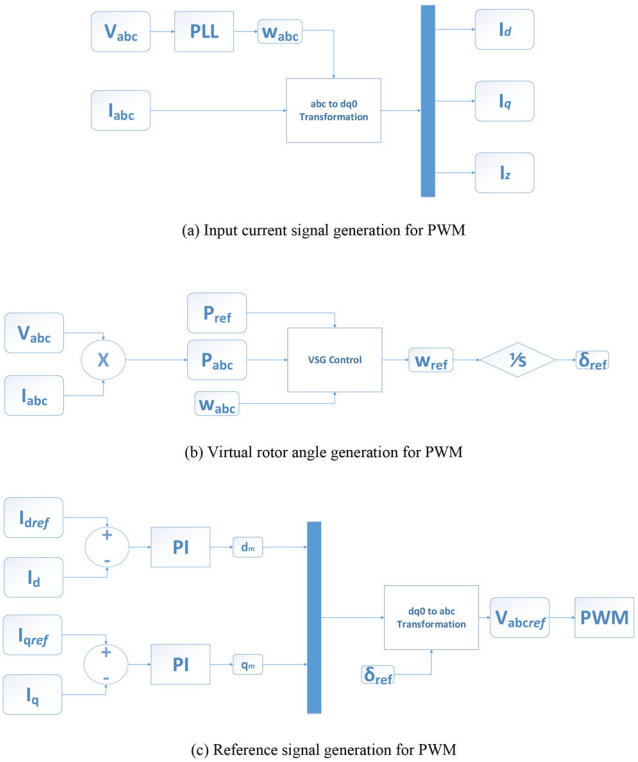


FIGURE 7. The generic diagram of the VSG. (a) Input current signal generation for PWM. (b) Virtual rotor angle generation for PWM. (c) Reference signal generation for PWM.

compares the instantaneous system frequency and ideal frequency for ensuring power balance by taking advantage of V2G connection. The VSG is performed by adjusting PWM belonging to the converter between the Charger Point and the MG. In that process, the VSG controls the voltage of DC bus of the converter for balancing power flow between the Charger Point and the MG.

3. TEST AND RESULTS

3.1. Input Data

The power system comprises MPPT-based solar panels, which can supply ~0.5 MVAr at 1000 W/m^2 and 25 °C, DFIG-based wind turbines, which can supply around 1.6 MVAr at 15 m/s wind speed, and a diesel synchronous generator on the supply side. On the demand side, there is a residential load area with a power demand of 2 MW. The paper proposes connecting a VSG mechanism with PEI to a V2G-based CS, which can operate as a load or a source depending on the system's frequency. The power generated by the diesel synchronous generator is analyzed to determine the optimal capacity of the emergency generator,

taking into account the VSG optimization rate and the volatility of environmental variables. Furthermore, four different case studies are examined to verify the effectiveness of the model as follows:

- ***Base case***: Frequency regulation by a conventional method.
- ***Case-1***: Frequency regulation by the proposed method with 15 m/s wind speed for DFIG-based wind turbine, 1000 W/m^2 irradiation, and 25 °C temperature for MPPT-controlled solar panel, 400 A current limitation for charger.
- ***Case-2***: Frequency regulation by the proposed method with 15 m/s wind speed for DFIG-based wind turbine, 1000 W/m^2 irradiation, and 25 °C temperature for MPPT-controlled solar panel, 350 A current limitation for charger.
- ***Case-3***: Frequency regulation by the proposed method with 15 m/s wind speed for DFIG-based wind turbine, 300 W/m^2 irradiation, and 25 °C temperature for MPPT-controlled solar panel, 400 A current limitation for charger.
- ***Case-4***: Frequency regulation by the proposed method with 10 m/s wind speed for DFIG-based wind turbine, 300 W/m^2 irradiation, and 25 °C temperature for MPPT-controlled solar panel, 400 A current limitation for charger.

3.2. Simulation and Results

The study presented in this article is based on a model that is simulated with a high level of precision (1/200,000) using the MATLAB and Simulink environment. The proposed control methods are applied to the model, and the results are analyzed under certain predetermined scenarios that reflect real-world trends and possible sources of uncertainty. The base case, which serves as a reference point for comparison, involves the use of conventional methods to compensate for frequency fluctuations.

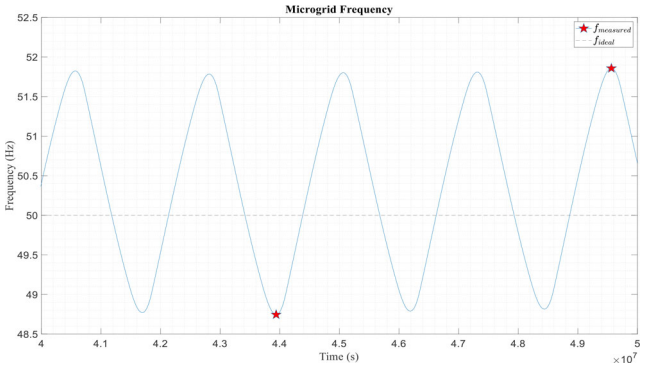


FIGURE 8. The response of system frequency in base case.

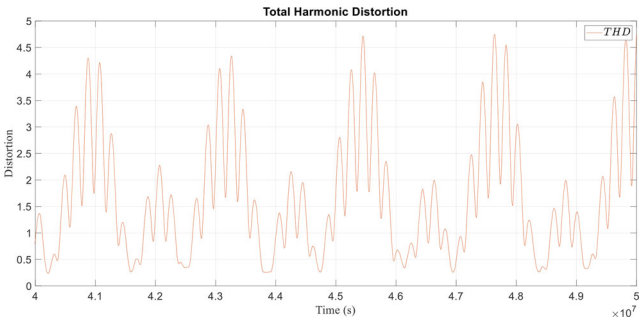
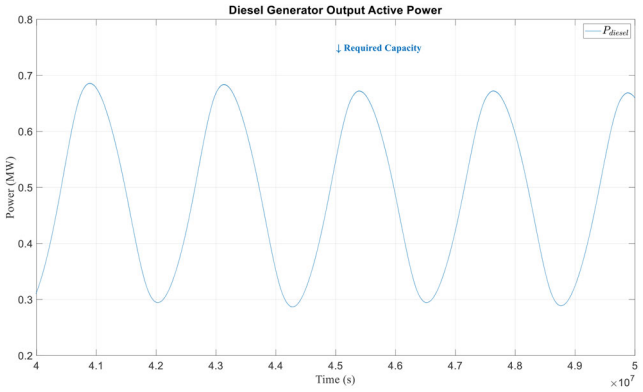
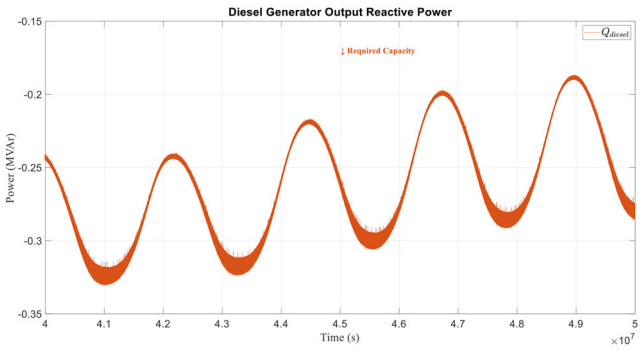


FIGURE 9. The total harmonic distortion of the system in base case.



a) The active power output of the DFIG



b) The reactive power output of the DFIG

FIGURE 10. The active and reactive power output of the DFIG in base case. (a) The active power output of the DFIG. (b) The reactive power output of the DFIG.

As mentioned previously, the proposed MG ensures its stability using conventional methods in the base case. In this case, the supply side is composed of MPPT-based solar panels operating at 1000 W/m^2 and 25 °C, DFIG-based wind turbines continuously spinning at 15 m/s, and an emergency synchronous generator, while the demand

side is assumed to be a residential area with a power demand of 2 MW.

In the base case, the aim is to maintain system stability using conventional methods. However, the system is unable to maintain frequency stability, as shown in Figure 8, and it is observed that the highest and lowest frequency values are 51.8 and 48.7 Hz, respectively. However, it is observed that wind turbines based on DFIGs exhibit harmonic distortion in the output side in Figure 9. Hence, as illustrated in Figure 10, the fluctuating output power of the 0.7605 MVAr emergency generator is unable to maintain the frequency within the desired range.

In Case-1, both the conventional method and VSG mechanism are simulated for FC implementation. Consequently, the VSG controlled by the CS is interconnected to the MG. The battery terminal voltage of the CS is set at 400 V and is integrated into the MG through the PEI. The system frequency in Case-1 is depicted in Figure 11. According to the results, the VSG mechanism and the emergency generator, with a capacity of ∼0.3420 MVAr and assumed fleet size of EVs, are able to keep the frequency fluctuation within the allowable limits, with the highest and lowest frequency points around 50.05 Hz.

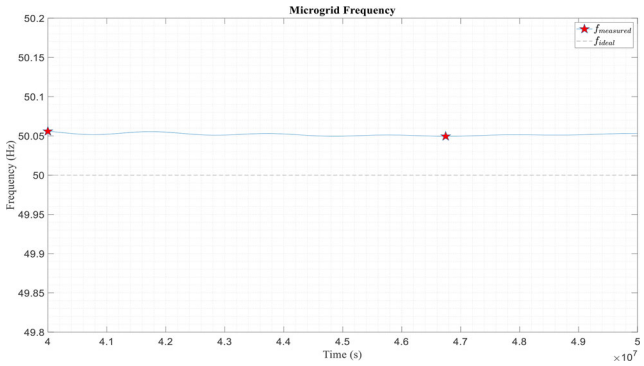


FIGURE 11. The response of system frequency in Case-1.

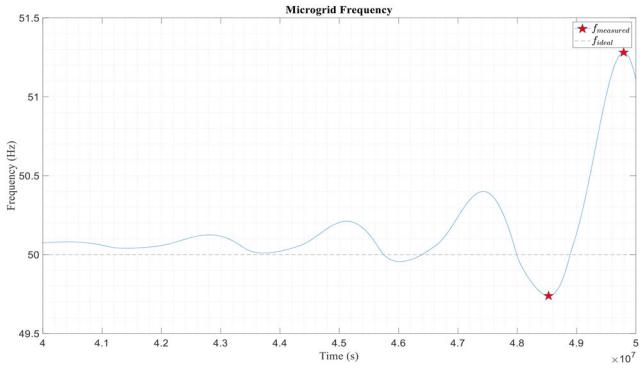


FIGURE 13. The response of system frequency in Case-2.

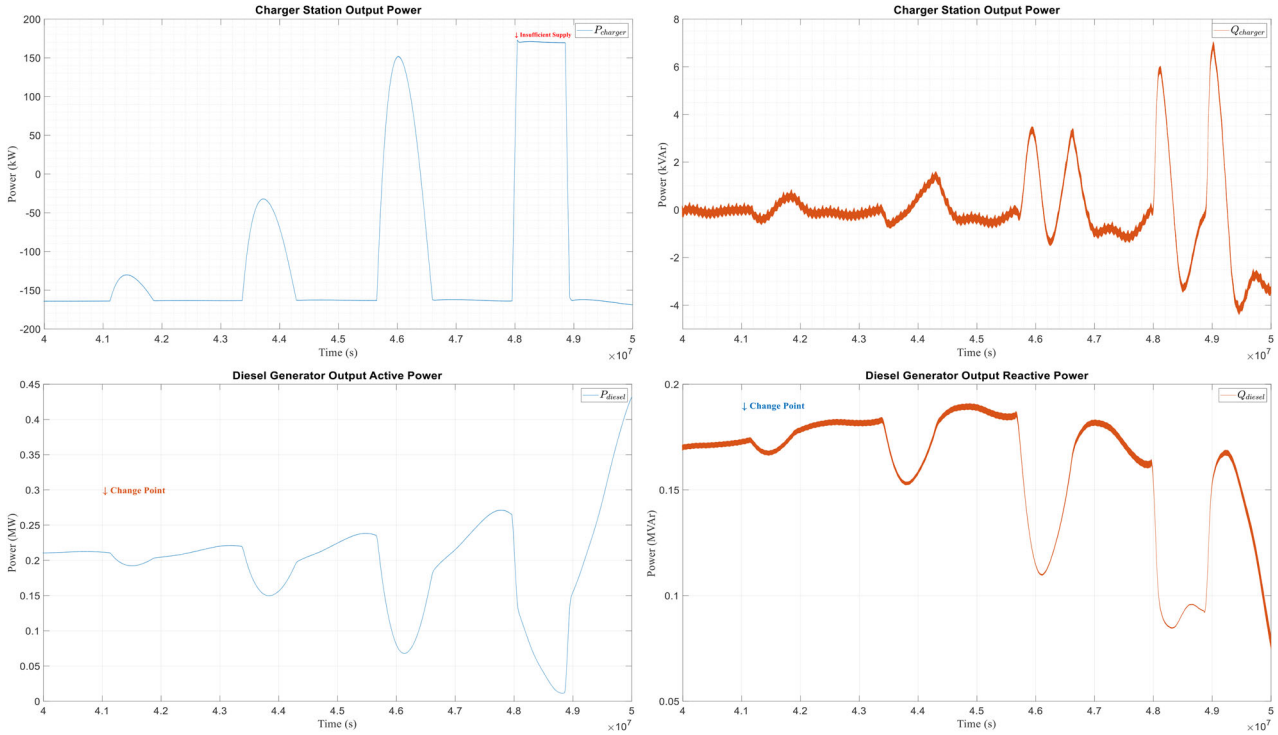


FIGURE 12. The output powers of the charging station and diesel generator in Case-2.

In Case-2, the fleet size of EVs is reduced to investigate its effect on the FC, and the maximum current capacity of the new fleet is decreased from 400 to 350 A in 4.1 s. The aim is to achieve system stability with a smaller fleet size of EVs as a result of this change. Figure 12 shows that the CS effectively injects and absorbs power for a certain time interval. However, at 4.8 s, the system faces a power mismatch due to an imbalance between the fleet size and demand, as well as the assumed smaller fleet size of EVs and the diesel generator with a capacity of ∼0.4609 MVAr. Thus, the effect of the fleet size of EVs is observable in an increase of maximum frequency from 50.05 to 51.28 Hz in the transition from Case-1 to Case-2, and the frequency fluctuation is measured in the range of 1.55 Hz in Case-2. As a result, the system cannot maintain its own stability due to lack of fleet size of EVs, as presented in Figure 13.

In Case-3, the performance of the proposed approach is observed under the conditions of 25 °C and a decrease in solar irradiation from 1000 to 300 W/m^2 at 4.1 s. Figure 14 shows that the system can maintain its own stability, with the frequency fluctuation occurring within allowable limits. Despite the RES-based DGs comprising MPPT-based solar panels with rapidly decreasing irradiation from 1000 to 300 W/m^2, it is observed that the CS and DFIG-based wind turbines respond effectively to the system based on

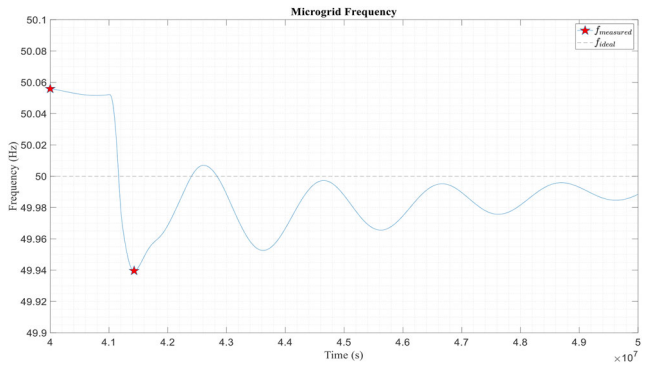


FIGURE 14. The response of system frequency in Case-3.

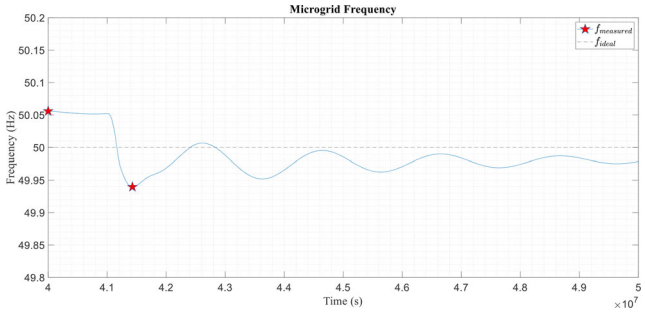


FIGURE 16. The response of system frequency in Case-4.

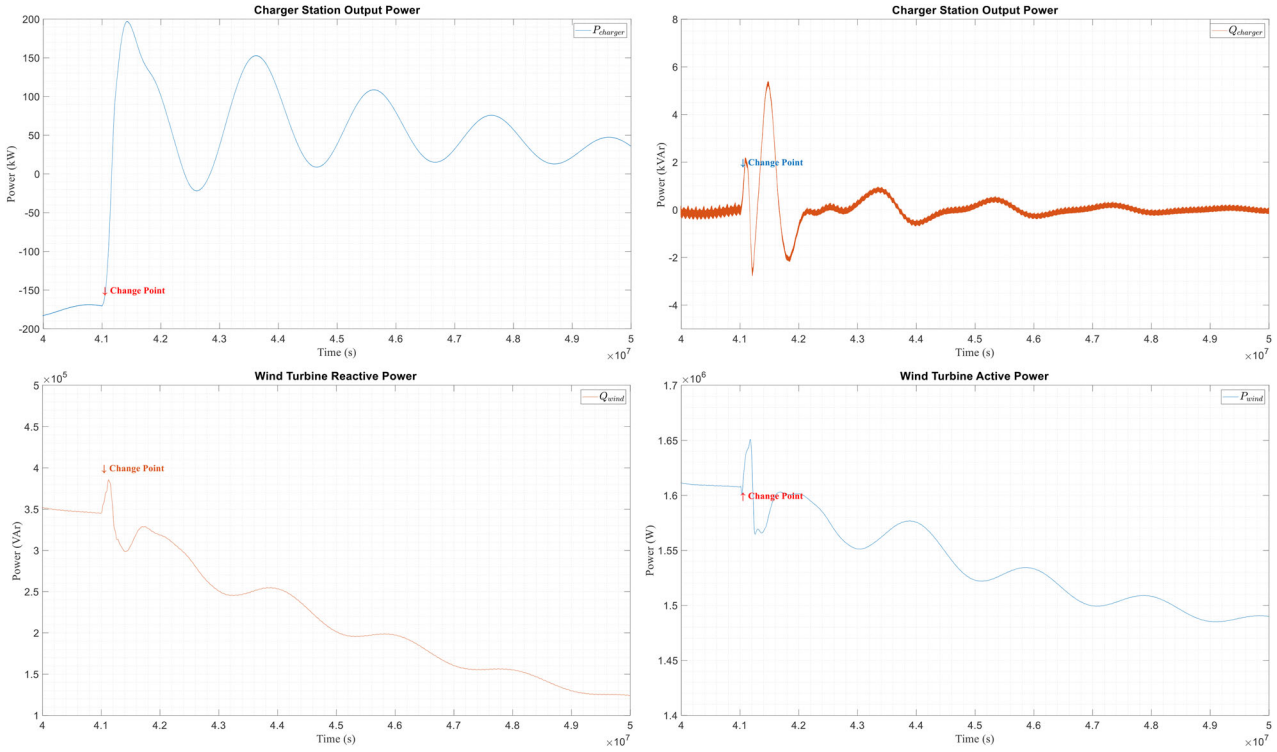


FIGURE 15. The output powers of the charging station and the DFIG in Case-3.

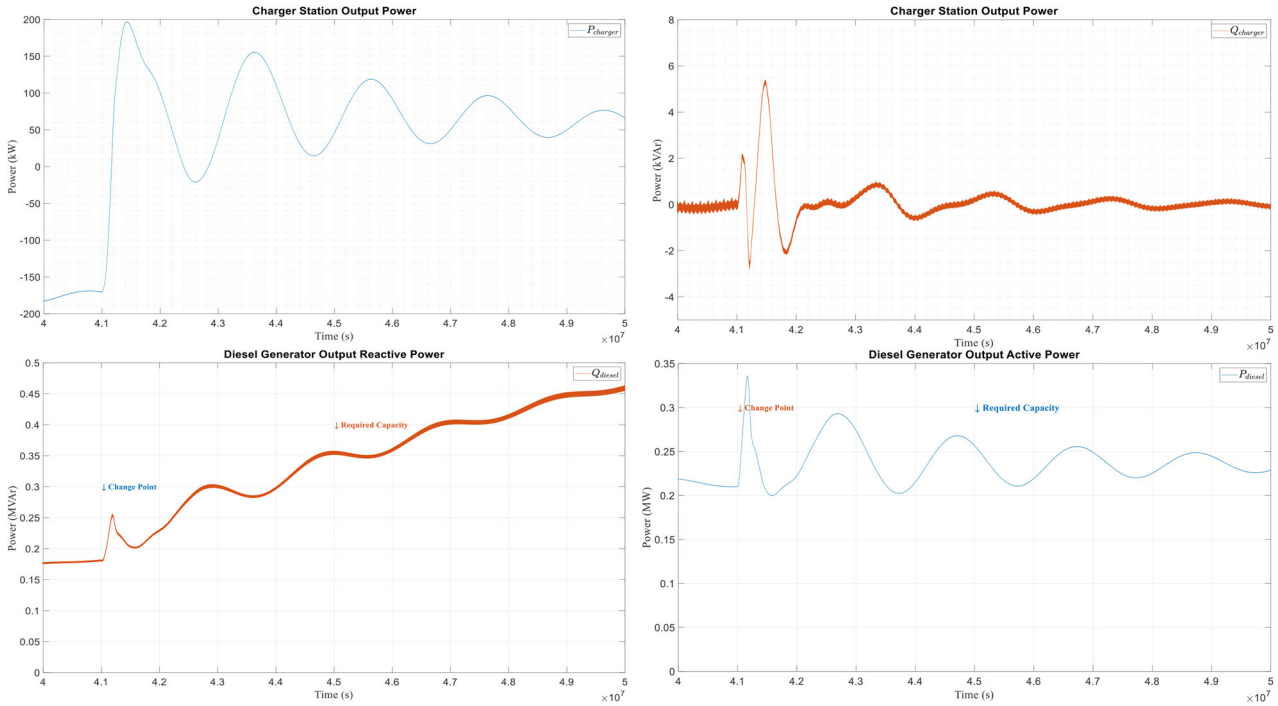


FIGURE 17. The output powers of the charging station and the diesel generator in Case-4.

Cases	Power (MVAr)
Base case	0.7605
Case-1	0.3420
Case-2	0.4609
Case-3	0.5280
Case-4	0.5521

TABLE 2. The size of the emergency diesel generator for each case.

the frequency change as shown in Figure 15. As a result of the uncertain behavior, the reactive power output of the diesel generator is calculated as 0.5280 MVAr with the assumed fleet size of EVs.

Finally, the behavior of the method under the condition of a decrease in solar irradiation from 1000 to 300 W/m^2 and wind speed from 15 to 10 m/s at 4.1 s is investigated in Case-4. Similar to Case-3, it is found that the frequency fluctuation is within permissible limits in Case-4, as shown in Figure 16. In this simulation, both the irradiation of the PV panels and the wind speed of the DFIG-based wind turbines decrease sharply from 1000 to 300 W/m^2 and from 15 to 10 m/s, respectively. Figure 17 demonstrates the output powers of the CS and diesel generator, and shows a sudden reversal in the CS's role depending on the change in system frequency at 4.1 s. In this regard, the capacity of the diesel generator is obtained as 0.5521 MVAr. The size of the emergency diesel generator for each case is shared in Table 2. The proposed system is tested under specific conditions that cause instability in the system. The simulation results show the difference between the conventional control system and the composite control system in terms of system stability, as seen in the base case and Case-2. The outputs are related to the impact of specific environmental variables and renewable production profiles, as shown in Case-3 and Case-4, which are compensated by the V2G mode of the VSG-controlled EVs. In these cases, it is observed that the contribution of the charger system and the emergency converter to the system is increased. This is also confirmed by the increasing role of the conventional system in the grid in Case-2, where a current limit is imposed on the charger system. The behavior of the system guides the determination of the size of system components to compensate for destabilizing effects under different limits.

4. CONCLUSION

The study was to provide a financial perspective for rescaling the capacity of emergency generators to keep the frequency of MGs stable in the case of instant changes on the supply and load side. In this direction, frequency characteristic was the main criterion, and various control and

configuration methods including MPPT, VSG, V2G, and DFIG were applied to the RES-based MGs.

The system was simulated on MATLAB and Simulink environment in the sense of frequency response depending on the changes in the RES and fleet size of EVs. The system was analyzed by starting with a conventional model in which only an emergency generator is used for compensating power imbalances, and then the response of the VSG mechanism to system stability was compared. In this direction, the study demonstrated the decrease in the range of frequency fluctuation from 3.1 to <0.05 Hz, in the peak point of frequency from 51.8 to 50.05 Hz, and in the capacity of EG from 0.7605 to 0.3420 MVAr with the addition of the VSG mechanism to the MG compared to the only conventional method. In addition, while the effect of decreasing solar irradiation and wind speed was noticed in the increase of the capacity of the EG from 0.3420 to 0.5521 MVAr, the effect of decreasing the size of the EV fleet was observed in the increase of the capacity of the emergency generator from 0.3420 to 0.4609 MVAr. In the scaling of MGs, the impacts of load side changes will be considered as the main area of interest in future studies.

DISCLOSURE STATEMENT

No potential conflict of interest was reported by the author(s).

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